3D Segregation of cell-like active particles

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Abstract

Segregation of cells of the same type from a mixture of different kinds of cells is a complex spatio-temporal process that is expected to play an important role in several pattern formation phenomena in life. The geometry and dynamics of segregating cells have extensively been studied in two dimensions (e.g., along surfaces), however, much less is known about the related behaviour in the 3-dimensional case which is obviously also very relevant in many practical situations. Motivated in part by recent experimental results, here we consider the adhesion difference-driven segregation of actively moving units, a fundamental but still poorly explored aspect of collective cell motility. We propose a model, which makes use of a recently published important result demonstrating that the interaction of tissue cells can be well approximated by an effective potential between them. This approach allows us to simulate the cell segregation process in three dimensions: a process that has been numerically extremely challenging and accomplished only in a few works. We assume a simple form of the interaction potential corresponding to hard-sphere type repulsive and softer, but also short-range adhesive forces. Our optimized numerical technique allows the simulation of several thousand cell-like particles over the whole pattern formation event. Our particles have a tendency to adhere through a mechanism that makes them both stay in touch and synchronize their direction of motion—but the interaction is limited to particles of the same kind. The results of the simulations are compared with the experimentally observed quantities and a reasonable agreement is found. In particular, we find an anomalously fast growth of the characteristic sizes of the patterns for intermediate time scales. The relevance of collectively moving clusters within the aggregates will be a subject of larger-scale calculations.

Keywords: segregation, three-dimensional, active matter, adhesion

1 Introduction

As an evergreen topic, the various forms of cell motility attract considerable attention in life sciences owing to their involvement in important biological fields such as embryonic development, wound healing and cancer progression [1], [2], [3], [4]. In all these fields of cell motility, a crucial aspect is the contribution of collective motion [5], [6], i.e. when cells move in a coordinated fashion and due to the collective effects they can be more effective, compared to cells moving as singles. The perspectives of collective motion in "active matter" research further enhance its popularity.

Over the years, the collective motion aspects have been studied both experimentally [7], [2], [3], [1], [8], [9], [10], and also in simulation platforms [11], [12], [13], [14], [15], [16], [17], while the main emphasis was on 2-dimensional systems offering less complicated access compared to the extra difficulties of 3D systems. However, the majority of real-life biological systems are 3D, making them more important though more challenging to study.

The difficulties of experimentally studying 3D collective motion are often apparent but it is equally difficult to study it through simulations. For example, it is possible to study 2D cell segregation by the Potts model approach [11], [15], while the application of this approach in 3D would be way more complex, and as yet we have not been aware of such application with success. The usefulness of simulation studies lies in the fact that by changing the parameters of models it is easier to identify which parameters play the decisive roles. Such changes in the parameters of live experimental systems rather than models are often more difficult if not unfeasible at all.

It is an outstandingly important question whether collective migration is present in and contributes to cell segregation phenomena. Here, we attempt to interpret the contribution of collective migration by analyzing the time dependency of segregated domain sizes during 3D segregation. This paper is based on the findings of two important earlier research fields: i) live cell experiments on 3D segregation; ii) modeling approaches describing the cell-cell interactions, i.e. forces, by an effective mechanical potential.

Cells' segregation in 3D was studied in several experimental works [18], [19], [20]. Considering the long-term behavior of 3D tissues as being similar to viscous fluids eventually led to finding analogies between the segregation of cell types and segregation of non-mixing fluid types, driven by the components' different surface tensions [21]. Several works studied the molecular aspects of heterotypic cell segregation trying to explain the final spatial configurations of segregated homotypic domains, while others considered the biomechanical aspects, such as the role of cortical tension regulation in the generation of tissue surface tension, the main drive of segregation [22], [21], [23], [24], [25], [26], [27], [28], [29]. For characterizing the dynamics of segregation, the time-dependent changes in linear sizes of the segregated domains were measured in various live cell as well as inorganic systems [10], [30], [21].

Cell-cell adhesion forces exerted by cell adhesion molecules of contacting cells and acting towards increasing the contact area can be approximated as simple attractive forces. Cortical tension, the force exerted by the contractile actomyosin network of cells' cortical cytoskeleton, acts towards decreasing the contact area of contacting cells and therefore it can be approximated as repulsive force. The summation of these two antagonizing forces depends on the distance of contacting cells and therefore it can be described as pairwise effective mechanical potential, similarly to the Lennard-Jones potential generally applied to intermolecular interactions [31].

2 The Mathematical Model: Collective Cell Migration Model

Our goal is to create a model in which particles move collectively. We employ and extend to 3D the model described earlier [32].

Owing to the biophysical properties of cells we are in the over-damped limit. We get relation between velocity and force as:

$$\vec{v}_i(t) = \frac{d\vec{r}_i(t)}{dt} = v_0\hat{n}_i(t) + \vec{F}_i(t) \tag{1}$$

Here, $\vec{r}_i(t)$ and $\vec{v}_i(t)$ is the position and velocity of the ith particle, respectively. The cells are treated as spherical particles and thus, position and velocity are given for the centre of mass of the particle. Furthermore, $\hat{n}_i(t)$ is unit vector of the heading direction of the ith particle and v_0 is magnitude of self-propulsion velocity, thus, $v_0\hat{n}_i$ ensures that the particle is "active" i.e. it moves even without any external driving force. $\vec{F}_i(t)$ is the external force experienced by the particle.
We are modeling cells as active particles that follow equation 1. That is the cells, or particles, are impacted by two factors: their own activity which is captured by the $v_0\hat{n}_i$ term and external forces, real and/or behavioral. In case of no external forces, the particle would continue to move in its heading direction, like a self-propelled particle.

The rule for change of heading direction is given by Equation 2 :

$$\frac{d\hat{n}_i(t)}{dt} = \frac{\hat{v}_i(t) - \hat{n}_i(t)}{\gamma} + \vec{\xi}_i \tag{2}$$

Here, $\hat{v}_i$ is the unit vector of velocity; and $\vec{\xi}_i$ is a vector representing noise, sampled from a sphere of radius $\xi/2$ centred at 0. The variable γ is the characteristic time scale associated with direction change.
Our modeled cells also change their heading directions, as is shows in equation 2. They tend to move towards the direction of the force they are being pushed in, which is the role of $\hat{v}_i$ in the equation. Biological systems cannot instantly change their directions, and are associated with a characteristic time scale γ. Particle would relax instantly as $\gamma \to 0$. Additionally, there is a noise term $\vec{\xi}_i$ which takes into account the

imperfect steering of cells and is a general property of the biological systems.

Equation 2, implies that the particles are moving collectively. Because a particle is responding to the difference between its own heading direction and the velocity it is experiencing. The velocity the particle is experiencing is a combination of its internal heading direction and the external driving force, $\vec{F_i}$ which includes contribution from other cells and boundary. So when a particle tries to align to this velocity it is also trying to align to the driving force and thus its neighbouring particles. This is similar to the models in which a particle moves in the average direction of its neighbours [33]. A crucial thing to note here is that as $\gamma \to \infty$ our collective of particles will behave like Active Brownian particles.

The external force, $\vec{F_i}$, in Equation 1 originates from two types of interactions—cell-cell and cell-wall.

$$\vec{F_i} = \sum_{j=1}^{N} \vec{F}^{cell}(\vec{r_i}, \vec{r_j}) + \vec{F}^{wall}(\vec{r_i}) \tag{3}$$

The cell-cell interactions, F^{cell}, can be attractive or repulsive depending on the distance, whereas the magnitude of this force depends not only on the distance but also on the type of interacting cells. Furthermore, in accordance with the observations we keep the forces short-ranged and weak.

Let's represent the distance between, the centres of, two particles as d_{ij}:

$$d_{ij} = ||\vec{r}_j - \vec{r}_i||_2$$

Force by jth particle on ith particle is given by :

$$\vec{F}^{cell}(\vec{r_i}, \vec{r_j}) = \frac{\vec{r_j} - \vec{r_i}}{d_{ij}} \begin{cases} C_{rep} \frac{d_{ij} - D_{eq}}{D_{eq}} & d_{ij} < D_{eq} \\ C_{adh} \frac{d_{ij} - D_{eq}}{D_0 - D_{eq}} & D_{eq} < d_{ij} < D_0 \\ 0 & D_0 < d_{ij} \end{cases} \tag{4}$$

Here, C_{rep} and C_{adh} are the maximum magnitude of repulsive and adhesive forces between two particles, respectively. D_{eq} and D_0 are defined as:

$$D_{eq} = R_{eq,i} + R_{eq,j}$$

$$D_0 = R_{0,i} + R_{0,j}$$

Here $R_{eq,i}$ is the radius of the ith particle, and $R_{0,i}$ is the interaction radius of the ith particle that is the if the edge of another particle is $R_{0,i}$ distance away from the center of particle i then there would be no force on particle i from it. Additionally, this also implies that $R_{0,i} \geq R_{eq,i}$.
Then, D_{eq} is the sum of radii of two interacting particles i and j. D_0 is the sum of interaction radii i.e. radius beyond which there will be no interaction between two

particles, of particles i and j.
Thus, if the distance between two particles is less than D_0 but greater than D_{eq} radii combined then they experience attraction, and if the distance between two particles is less than D_{eq} combined then they experience a repulsion. The curve for pair-force looks like a linear curve, which changes sign at D_{eq}. This should tend to make homotypic particles on average maintain a distance of their diameters between their centers. And since all the forces are weak this conditions would be violated frequently, which helps model the soft nature of cells, and cells can interact more freely. This is seen in the simulations too, if we increase the magnitude of repulsion, the system ceases to produce geometries and instead settles in a local energy minima, resembling solid crystals.

We also assume that there can be no attraction between the particles and their surrounding media. This is also how the experiments were conducted, there wasn't a possibility of cell-substrate adhesion. A particle experiences only repulsive interactions with the boundary or wall, F^{wall}. This helps keep the particle bounded within the simulation arena.

Let, d_{iw} be distance of particle i from the wall.

$$d_{iw} = ||\vec{r}_{wall} - \vec{r}_i||_2$$

Force from the wall on particle i is:

$$\vec{e}_{iw} = \frac{\vec{r}_{wall} - \vec{r}_i}{d_{iw}}$$

$$\vec{F}^{wall}(\vec{r}_i) = -\vec{e}_{iw} \begin{cases} C_{wall} \exp\left(\frac{-2d_{iw}}{R_0}\right) & d_{iw} < R_0 \\ 0 & R_0 < d_{iw} \end{cases} \tag{5}$$

C_{wall} is the maximum force of repulsion a particle can experience from a cell-wall interaction.
If the distance between particle i center and the wall is less than R_0 then particle experiences a strong repulsion from the wall.

Since our system of interest contains cells of two types: red(R) and green(G), we apply our model to binary mixture of particles. So many of the parameters have different values for red particles and green particles. They are:

- Self-propulsion velocity: Typically $v_{0,R} \geq v_{0,G}$
- Size of particle: $R_{eq,R} > R_{eq,G}$
- Size of attraction sphere: $R_{0,R} > R_{0,G}$
- Magnitude of attraction: $C_{adh}^{R-R} > C_{adh}^{G-G}$

Thus, the red particles are larger, more motile and have stronger adhesive force among each other. Additionally, there is also also a force of attraction between the red and

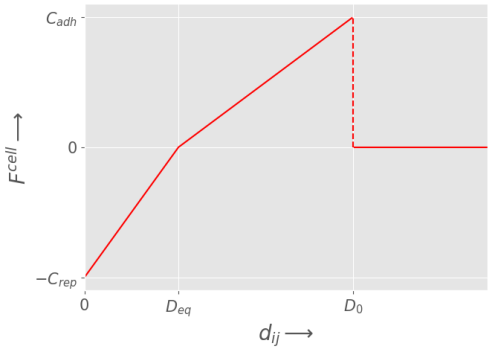


Fig. 1: **Force vs distance** Curve from Equation 4. The variables are same as that for the equation.

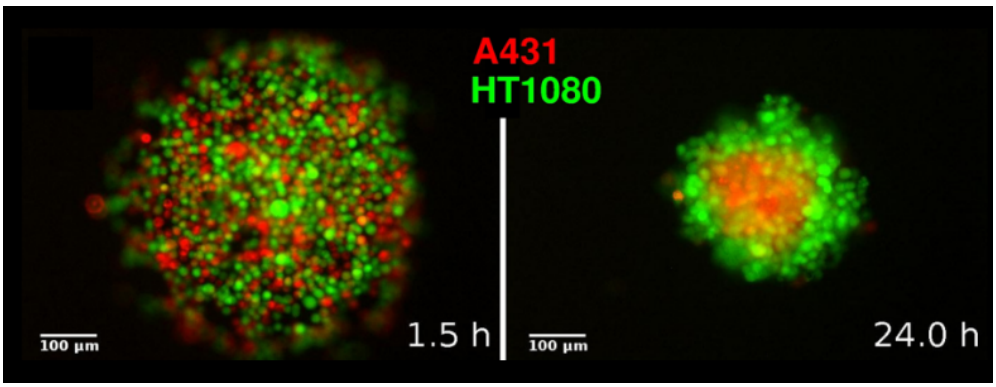
green particles, C_{adh}^{R-G}. It is related to other forces of attraction as:

$$C_{adh}^{R-R} > C_{adh}^{G-G} > C_{adh}^{R-G}$$

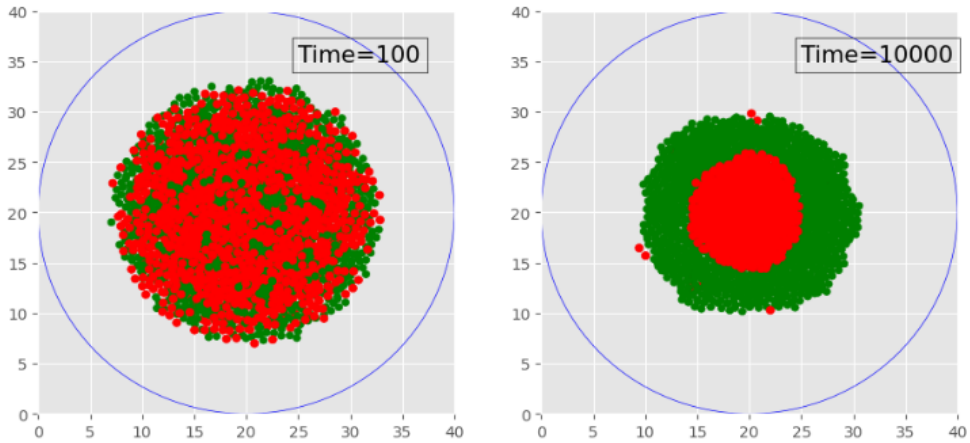
3 Simulation Results

Simulating the equations mentioned in section 2, from the numerical approaches mentioned in section 4, we simulated systems with varied configurations of parameter values. Here again, we would like to talk about the experimental results that motivated our exploration[30], with the permission of the authors. In the experiments, single-cell suspensions of a binary cell mixture of approximately equal number of cells were mixed in suspensions in non-adherent aggregation chamber wells with only cell-cell adhesion and no cell-substrate adhesion. Then they observed these suspensions for several hours until final spatial configurations of the segregated domains were reached. Their images are given in Figure 2 and 3. We have reproduced here two experiments here. Figure 2a represents A431 human epithelial carcinoma cells and HT1080 human fibrosarcoma cells. Figure 3a represents ectoderm and mesoderm created from zebrafish, *Danio rerio*, progenitor cells. In both these plots, the cell type with red fluorescence has higher homotypic contact strength than the cells with green fluorescence. As can be clearly seen from the figure, the cells self-organized and segregated in geometrical patterns forming spheroids, with the cells having higher homotypic strength, that is, the cells with red fluorescence segregated inwards.

Similarly, the modeling scheme mentioned in the section 2 and the numerical method and parameter values mentioned in the section 4 can produce similar segregation results. See Figure 2 and 3. Like the experiments, the shape of the simulation arena is a cylinder with a hemispherical bottom. The cells are also about 3000 in number, with equal number of red and green particles, and mixed randomly at the start of the simulation. Majorly, like simulations the particles in the red have a higher magnitude



(a) 3D segregation of human A431 epithelial carcinoma (red) and human HT1080 fibrosarcoma (green) cells.

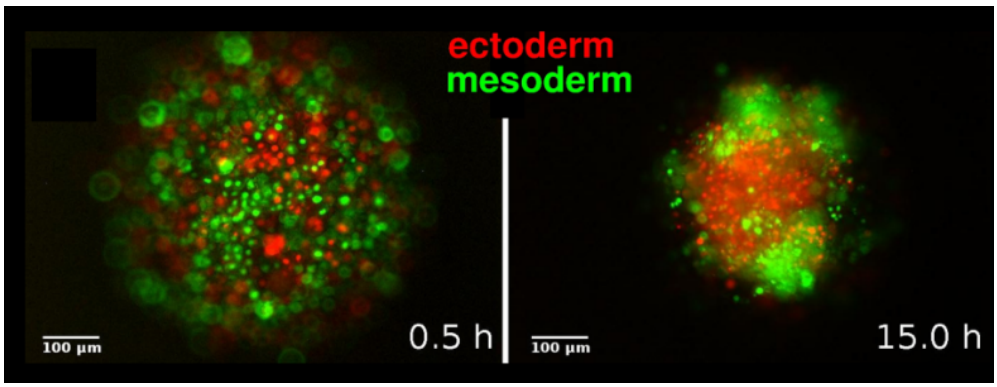


(b) 3D Segregation of simulation particles. Total number of particles is 3000. For red particles: $N_R = 1500$. For green particles: $N_G = 1500$. The blue ring is how the simulation well looks from top.

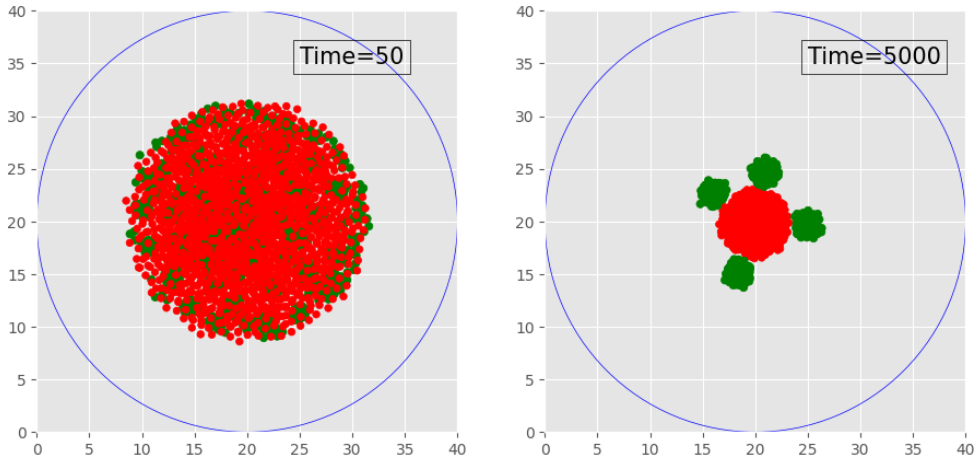
Fig. 2: **3D segregation observed in the experiments and simulations.** In each image: left panels show the initial phase of segregation, and the right panels show the final segregated spatial configuration; time(in hours and simulation time units, respectively) after the initial mixing of binary cell suspensions is indicated at lower right corners; a length scale is also provided.

of attraction than green particles i.e. $C_{adh}^{R-R} > C_{adh}^{R-G}$. In the figure, the images shown are only top views. For full view of the segregated image see Figure 4. Figure 6 shows how the final configuration changes with total number of simulation particles.

To quantify the behaviour of the system the measure of cluster size is used. See



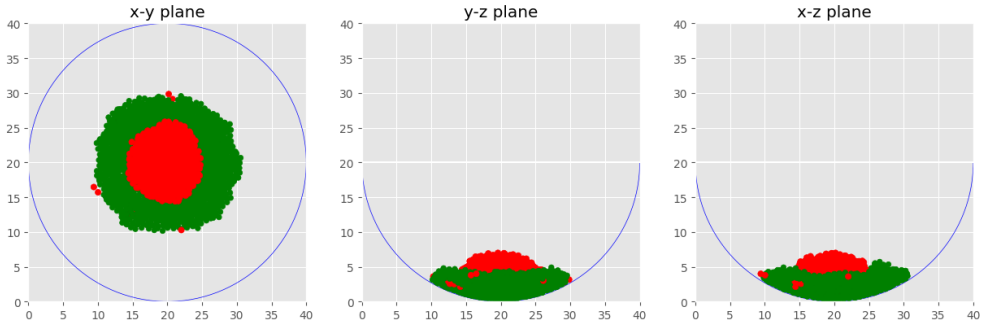
(a) Total number of particles is 3000. For red particles: $N_R = 1500$. For green particles: $N_G = 1500$.



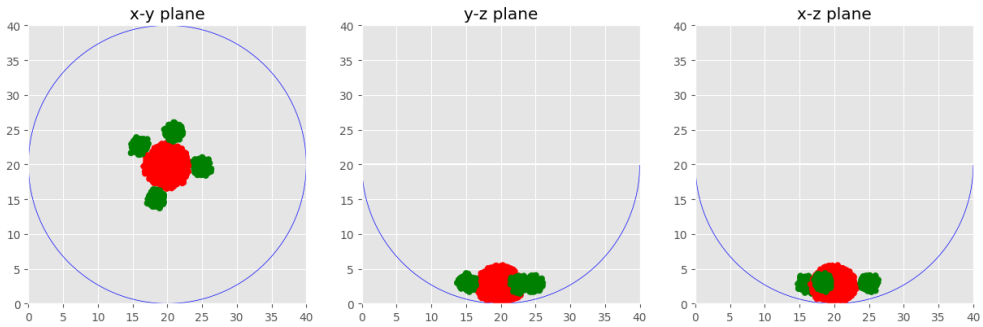
(b) Total number of particles is 3000. For red particles: $N_R = 1500$. For green particles: $N_G = 1500$.

Fig. 3: **3D segregation observed in the experiments and simulations.** In each image: left panels show the initial phase of segregation, and the right panels show the final segregated spatial configuration; time(in hours and simulation time units, respectively) after the initial mixing of binary cell suspensions is indicated at lower right corners; a length scale is also provided.

Figure 5. From the figures, the segregation starts slowly at first, then the growth rate speeds up and eventually the system stabilizes. Our simulations produced similar results. Even the scaling looks similar, if we consider that diameter of cells in experiments was about 10 microns, and diameter for simulations was about 1 simulation unit.



(a) Full 3D view of Figure 2b Time = 10000



(b) Full 3D view of Figure 3b Time = 5000

Fig. 4: **3D segregation simulation images in different planes** This image is corresponds to full 3D view of images belonging to Figure 2 and 3. Figure 4a corresponds to Figure 2b Time = 10000, and Figure 4b corresponds to Figure 3b Time = 5000.

4 Methods

4.1 The Numerical Model

Equation 1 can be interpreted as:

$$d\vec{r}_i(t) = (v_0 \cdot dt)\hat{n}_i(t) + \vec{F}_i(t)dt$$

Thus position would get updated as:

$$\vec{r}_i(t + dt) = \vec{r}_i(t) + d\vec{r}_i(t) = (v_0 \cdot dt)\hat{n}_i(t) + \vec{F}_i(t)dt$$

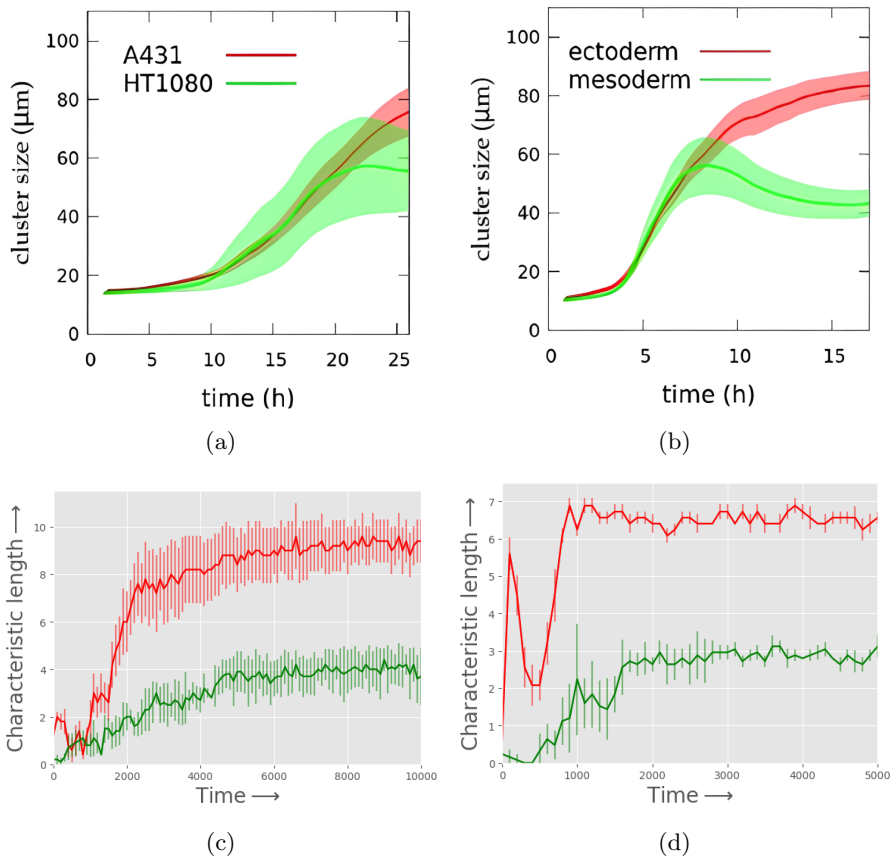


Fig. 5: **3D segregation cluster size observed in the experiments** 5a: Analysis of segregated cell cluster sizes in A431 keratinocyte and HT1080 fibrosarcoma mixtures from Figure 2a. 5c: Analysis of segregated particles characteristic length corresponding to Figure 2b.
Similarly, 5b: Analysis of segregated cell cluster sizes in zebrafish ectoderm and mesoderm mixtures from Figure 3a. 5d: Analysis of segregated particles characteristic length corresponding to Figure 3b.

The discretized equations would be:

$$r_x[i,t+1] = r_x[i,t] + (v_0 \hat{n}_x[i,t] + F_x[i,t]) \cdot dt$$

Here $r_x[i,t], r_y[i,t], r_z[i,t]$ represent the (x,y,z) coordinate value of the ith particle at the tth time step.

While $\hat{n}_x[i,t], \hat{n}_y[i,t], \hat{n}_z[i,t]$ represent the (x,y,z) component of the unit vector of the heading direction of the ith particle at the tth time step i.e. $\hat{n}_i(t)$.

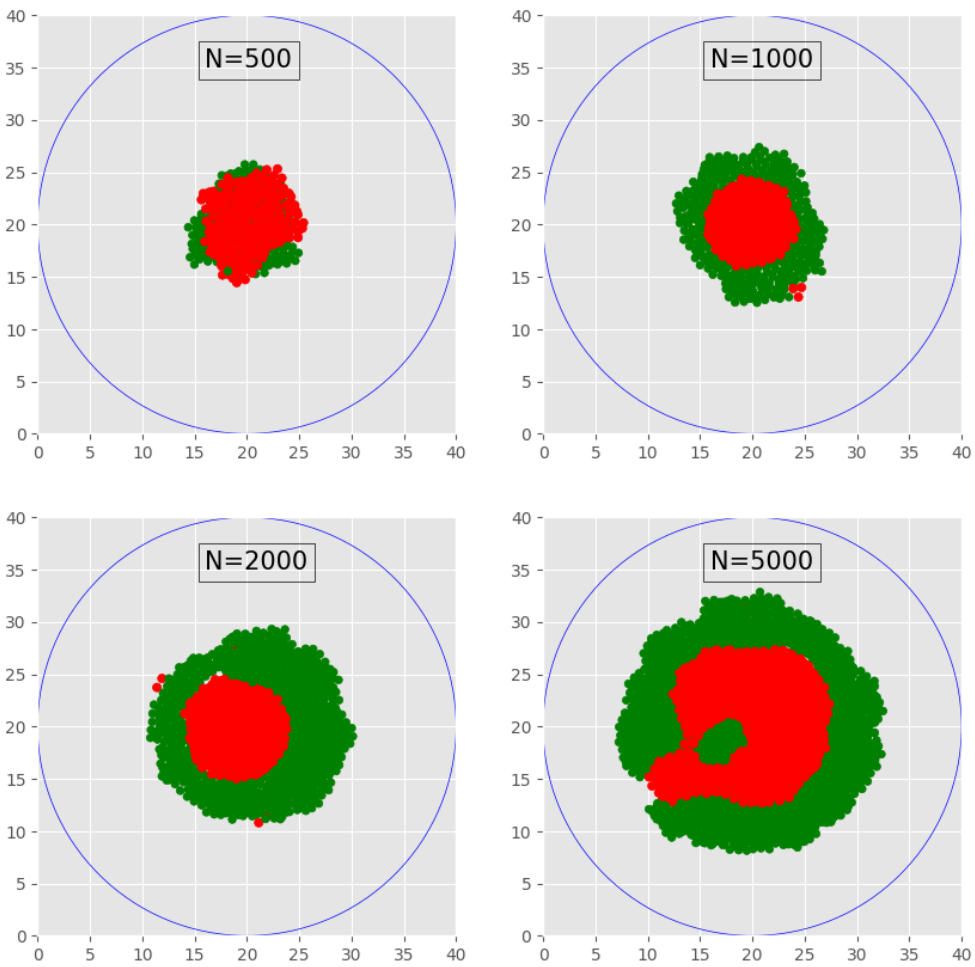


Fig. 6: **Final configuration for different particle numbers**. This is the top view of the final configuration. The number of particles in the simulation are written on each image. The simulation parameters here are the same as Figure 4a, except for numbers.

Similarly, $F_x[i,t], F_y[i,t], F_z[i,t]$ represent the (x,y,z) component of the unit vector of the external force of the ith particle at the tth time step i.e. $\vec{F_i}(t)$.

Equation 2 can be rewritten as:

$$d\hat{n}_i(t) = (\hat{v}_i(t) - \hat{n}_i(t))\frac{dt}{\gamma} + \vec{\xi}(t)dt$$

Here, the three components of the noise vector were generated by taking three random numbers independently from a uniform distribution U$[-\frac{\xi}{2}, \frac{\xi}{2}]$. If the magnitude of this vector is less than $\xi/2$, where ξ is our external parameter for noise, then it is accepted otherwise rejected.

The discretized equations, for updating heading directions, would be:

$$\hat{n}_x[i,t+1] = \hat{n}_x[i,t] + (\hat{v}_x[i,t] - \hat{n}_x[i,t])\frac{dt}{\gamma} + \xi_x dt$$

The $\hat{v}_x[i,t], \hat{v}_y[i,t], \hat{v}_z[i,t]$ is the (x,y,z) component of the unit vector of velocity at time of the ith particle at the tth time step i.e. $\vec{dr}_i(t)$. Explicitly, it looks like $\forall k \in \{x, y, z\}$

$$\hat{v}_k[i,t] = \frac{v_0\hat{n}_k[i,t] + F_k[i,t]}{\sqrt{(v_0\hat{n}_x[i,t] + F_x[i,t])^2 + (v_0\hat{n}_y[i,t] + F_y[i,t])^2 + (v_0\hat{n}_z[i,t] + F_z[i,t])^2}}$$

Similarly, $\hat{n}_k[i,t]$ is not the final value. Since it is a component of the unit vector, we need to normalise it. This looks like $\forall k \in \{x, y, z\}$:

$$\hat{n}_k[i,t+1] = \frac{\hat{n}_k[i,t+1]}{\sqrt{(\hat{n}_x[i,t+1])^2 + (\hat{n}_y[i,t+1])^2 + (\hat{n}_z[i,t+1])^2}}$$

4.2 The Simulation

The shape of the experimental setup, and also the shape of the simulation arena in which the particles are simulated, hereafter referred to as simulation well, is a cylinder of diameter L and height L/2 and a hemispherical bottom of diameter L.

Initially, the particles, typically with equal number of red and green particles, are randomly placed inside the simulation well and are made to fall with a weak gravitation and purely repulsive interactions. This gave an aggregate in which the particles are uniformly mixed and well settled at the bottom of the simulation well, at the start of the simulation. This is also how the experiments were performed.
At t=0, particles are provided random heading directions. After this the discretized equations stated in the section 4.1 are used and system is numerically integrated for a finite number of iterations, usually around $15000/dt$.

Additionally, the equations provided in the section 4.1 are incomplete as the equations are integrated using a 4th-order Runge-Kutta solver to improve the accuracy. Additionally, the discretized equations are written differently in the program as we have run simulations using parallel programming with GPUs has been employed to improve performance of the program.

4.3 Characteristic Length Calculation

To calculate the characteristic length for simulation structures, we first pick a slice of about three particle diameters thick. This slice is chosen such that it must contain the maximum number of particles, and even ratio of red to green particles. We use this slice to perform further statistics, this is similar to how data was gathered in the experiments for cluster size analysis (Figure 5). Then from the center of this slice, we calculate the amount, at distance r, of red and green color at that position. Basically, at that distance r we create a shell of width dr, and within that shell we have a mixture of red and green particles, we create a function that outputs the probablity of finding red particle within this mixture and substract from it the probability of finding green particle. So if the shell has only red particles our function returns +1, and for only green particles it returns -1, and for different mixtures we get an intermediate value. From this curve we calculate the characteristic length by finding the largest continuous regions with high amount of a particular color. This high amount is dictated by a threshold, which is +0.95 for red particles and -0.95 for green particles, which posit a 97.5% probability of finding red and green particles, respectively.

4.4 Parameters

Typical parameter values are given in Table 1.

5 Discussion

6 Conclusion

This paper highlights the importance of understanding 3D cell segregation, an essential process in various biological phenomena. The study aims to bridge the gap between two-dimensional models and the complexities of three-dimensional cellular interactions, which are pivotal in areas like embryonic development, wound healing, and cancer progression. The simulation results demonstrate significant insights into the behavior of cell-like particles in three dimensions. These results not only validate the proposed model but also offer a detailed understanding of the mechanisms driving cell segregation, showcasing patterns and dynamics that mirror biological realities. This research marks a significant advance in the simulation of 3D cell segregation. By developing a comprehensive mathematical model and employing robust simulation techniques, the study sheds light on the complex dynamics of cellular interactions in three dimensions. The findings enhance our understanding of cell motility and segregation.

Supplementary information.

Acknowledgments.

Declarations

Table 1: The set of parameters appearing in equations (1)-(5), along with their corresponding symbols and typical simulation values.

Parameter description	Name	Value
Number Red particles	N_R	10^3
Number Green particles	N_G	10^3
Red cell velocity	v_R	$10^{-2} - 10^{-1}$
Green cell velocity	v_G	$10^{-2} - 10^{-1}$
Cell-Cell repulsive force coeff.	C_{rep}	$10^{-1} - 10^0$
Cell-Wall repulsive force coeff.	C_{wall}	$10^{-1} - 10^0$
Red-Red adhesive force coef	C_{adh}^{R-R}	$10^{-1} - 10^0$
Green-Green adhesive force coeff.	C_{adh}^{G-G}	$10^{-1} - 10^0$
Red-Green adhesive force coeff.	C_{adh}^{R-G}	$10^{-1} - 10^0$
Noise	η	$10^{-1} - 10^0$
Red cell radius	$R_{eq,R}$	$10^{-1} - 10^0$
Green cell radius	$R_{eq,G}$	$10^{-1} - 10^0$
Red cell interaction radius	$R_{0,R}$	$10^{-1} - 10^0$
Green cell interaction radius	$R_{0,G}$	$10^{-1} - 10^0$
Linear system size	L	40
Relaxation time	γ	$10^{-1} - 10^1$

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